电磁学 Electromagentism *classical

Four Forces in nature

1. Strong force
2. Electromanegtic force
3. Weak force
4. Gravity

Classical EM (Electromagnetism) theory is a field theory.
$\mathbb{R}^3$ vector fields

Electric Field $\mathbf{E}$
Magentic Field $\mathbf{B}$
} ⇒ satisfy Maxwell's Equations

Electric Charges

- Plus or minus.
  - Electrons have $-e$. $(e>0)$
  - Protons have $+e$.
- Conserved Quantity
  → U(1) gauge symmetry*
- Quantized
  - Basic unit is $e \approx 1.60 \times 10^{-19}$ C

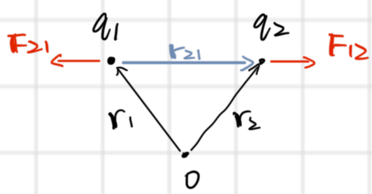
Electrostatics 静电学

- Superposition Principle 叠加原理.
  ⇒ Only need to study 2 charges.

  * In general the forces between charges are hard to derive.
  The force on Q from q, at t, relates to:
  ① velocity of q at time t_r (retarded time) $^* t_r = t - \frac{d(q_{t_r}, Q_t)}{c}$
  ② acceleration of q at t_r
  ③ velocity of Q at time t.

Now Assume everything is at rest.

- Coulomb's Law



$$\mathbf{F}_{12} = k \frac{q_1 q_2}{|\mathbf{r}_{21}|^2} \hat{\mathbf{r}}_{21}, \quad \hat{\alpha} := \frac{\alpha}{|\alpha|}.$$

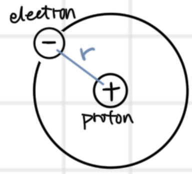
$$= \frac{k q_1 q_2 \mathbf{r}_{21}}{|\mathbf{r}_{21}|^3}$$

$k \approx 9 \times 10^9\ N \cdot m^2/C^2$
$= \frac{1}{4\pi\varepsilon_0}$,
$\varepsilon_0 \approx 8.85 \times 10^{-12}\ C^2/(N \cdot m^2)$
Permittivity of free space

$\mathbf{F}_{21} = -\mathbf{F}_{12}$ (Newton's 3rd Law)

If q_1 & q_2 have $\begin{cases} \text{the same sign} \Rightarrow \text{replusive} \\ \text{opposite sign} \Rightarrow \text{attractive} \end{cases}$

Example. Hydrogen atom



$r \approx 5.3 \times 10^{-11} m$

$F_c = k \cdot \frac{e^2}{r^2} \approx \boxed{8.2 \times 10^{-8} N}$

Corresponding Gravity force:

$m_e \approx 9.11 \times 10^{-31} kg$, $m_p \approx 1.67 \times 10^{-27} kg$

$F_G = G \cdot \frac{m_e m_p}{r^2} \approx \boxed{3.6 \times 10^{-47} N}$

$F_G << F_C$

· Electric Fields

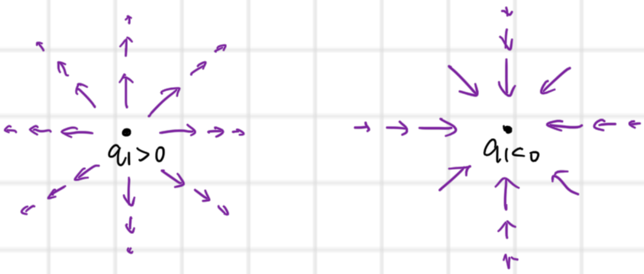
For n charges and a test charge q_0,

$\mathbf{F}_0 = q_0 \mathbf{E}$, $\quad \mathbf{E} = \sum_i \frac{k q_i}{\mathbf{r}_{0i}^2} \hat{\mathbf{r}}_{0i}$

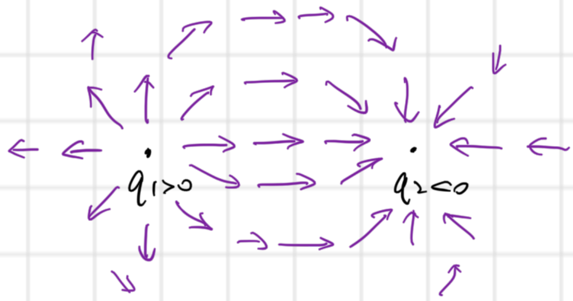
$\mathbf{E}(\mathbf{r}_0)$ is called electric field created by $q_1, \cdots, q_N$ at $\mathbf{r}_0$

Vector Field in $\mathbb{R}^3$

• $N = 1$



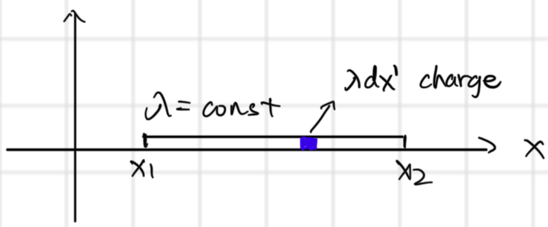
• $N = 2$



Continous Charge Disturbutions

• Volume charge Density
  $\rho(x, y, z)$ 3D
• Surface charge Density
  $\sigma(x, y)$ 2D
• Linear charge Density
  $\lambda(x)$ 1D

Ex. Charged rod.



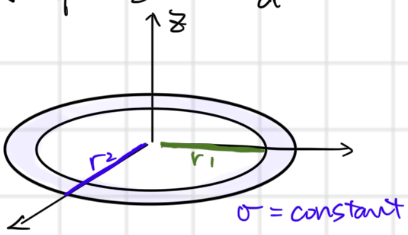
$$\mathbf{E}(x, y, z) = \int_{x_1}^{x_2} \frac{k \cdot \lambda dx' \cdot (x - x', y, z)'}{[(x - x')^2 + y^2 + z^2]^{3/2}}$$

For infinitely long rod, $\mathbf{E}_x = 0$,

$\mathbf{E}_y = \frac{2k\lambda y}{y^2 + z^2}$, $\mathbf{E}_z = \frac{2k\lambda z}{y^2 + z^2}$,

$|E| = \sqrt{E_y^2 + E_z^2} = \frac{2k\lambda}{d^2}$.

Ex.



$$E(0,0,z) = \int_0^{2\pi}\int_{r_1}^{r_2} \frac{k\sigma r\,dr\,d\theta}{[(r\cos\theta)^2+(r\sin\theta)^2+z^2]^{3/2}} \cdot \begin{pmatrix}-r\cos\theta \\ -r\sin\theta \\ z\end{pmatrix}$$

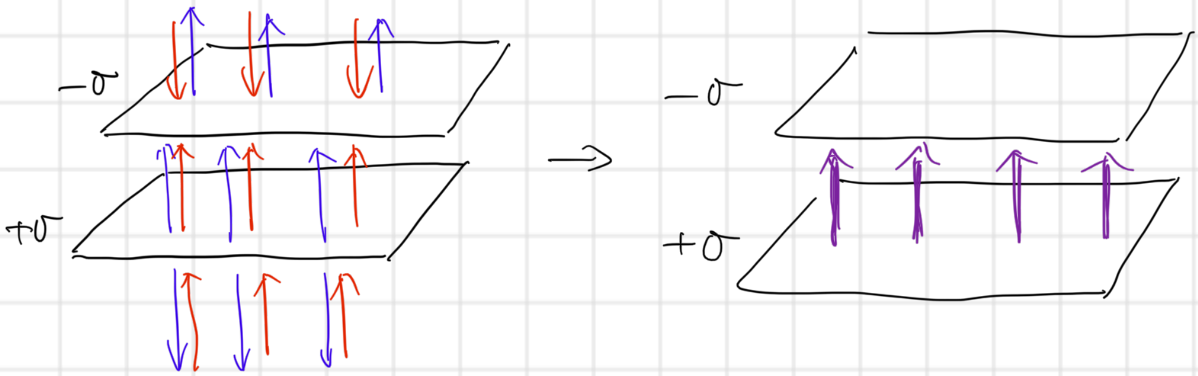
$$E_z(z) = \int_{r_1}^{r_2} \frac{k\sigma 2\pi r\,dr \cdot z}{[r^2+z^2]^{3/2}}$$

$$= 2\pi\sigma k z\left[\frac{1}{\sqrt{r_1^2+z^2}} - \frac{1}{\sqrt{r_2^2+z^2}}\right]$$

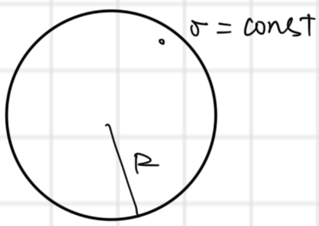
Infinite plane of charge: $r_1 \to 0$, $r_2 \to +\infty$.

$$E_z(z) = 2\pi\sigma k \cdot \frac{z}{|z|} = (\operatorname{sgn} z) \cdot \frac{\sigma}{2\varepsilon_0}$$

Ex. Parallel planes of charge.



Ex. Spherical shell of charge



$$E(r) = \begin{cases} 0, & r < R \\ k \cdot \frac{4\pi R^2 \sigma}{r^2}\hat{r}, & r > R \end{cases}$$